

CS471 – Lesson 22 In-Class Worksheet

Dynamic Probabilistic Systems: Markov Models

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Represent dynamic systems as transition matrices, state and evaluate the core Markov assumptions, and compute multi-step state probabilities.

1. Theater Weather State Transition Model

Consider an autonomous strike planning system with three discrete weather states:

$$S = \{\text{Clear } (C), \text{Cloudy } (K), \text{Storm } (R)\}$$

The single-step transition probabilities are given by:

- From Clear: 70% stay Clear, 10% transition to Cloudy, 20% transition to Storm.
- From Cloudy: 30% transition to Clear, 60% stay Cloudy, 10% transition to Storm.
- From Storm: 10% transition to Clear, 20% transition to Cloudy, 70% stay Storm.

a. Construct the 3×3 stochastic transition matrix T , verifying that each row sums to 1.0.

b. Telemetry confirms that on Day 0 the weather is Clear with 100% certainty:

$$\mathbf{p}_0 = \langle P(C) = 1.0, \quad P(K) = 0.0, \quad P(R) = 0.0 \rangle$$

Compute the state probability distribution vector on Day 1 ($\mathbf{p}_1$).

c. Using the mini-forward marginalization formula, compute the probability that the weather is Clear on Day 2 ($P(X_2 = C)$). Show all three component terms.

2. Assumptions of Markov Models

- a. State the **First-Order Markov Assumption** mathematically and explain its operational meaning for an autonomous mission planner.

- b. Explain the **Stationary Assumption (Time-Homogeneity)**. Describe an operational military scenario where this assumption is violated.

Part 2: Student Independent Practice

Objective: Independently compute multi-step state distributions, evaluate temporal trajectories, and derive stationary distributions.

Tactical Satellite Communications (SATCOM) Degradation

A Space Force communications satellite experiences dynamic atmospheric and cyber jamming. The satellite operates in one of three states:

$$S = \{\text{Operational } (O), \text{Degraded } (D), \text{Severed } (S)\}$$

The transition matrix across hourly intervals is:

$$T = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.20 & 0.70 & 0.10 \\ 0.00 & 0.10 & 0.90 \end{pmatrix}$$

where rows and columns are ordered $\langle O, D, S \rangle$.

1. At $t = 0$, the link is confirmed to be fully **Operational** ($\mathbf{p}_0 = \langle 1.0, 0.0, 0.0 \rangle$).

a. Compute the probability distribution at $t = 1$: $\mathbf{p}_1$.

b. Compute the probability that the link is fully **Operational** at time $t = 2$: $P(X_2 = O)$.

c. Compute the probability that the link is completely **Severed** at time $t = 2$: $P(X_2 = S)$.

2. **Trajectory Evaluation:** Compute the joint probability of the exact temporal sequence where the link starts Operational at $t = 0$, becomes Degraded at $t = 1$, and is Severed at $t = 2$:

$$P(X_0 = O, X_1 = D, X_2 = S)$$

- 3. Absorbing vs. Ergodic Behavior:** Row 3 of T is $(0.00 \ 0.10 \ 0.90)$. Can the satellite ever re-establish an Operational link (O) once it enters the Severed state (S)? Justify your answer.